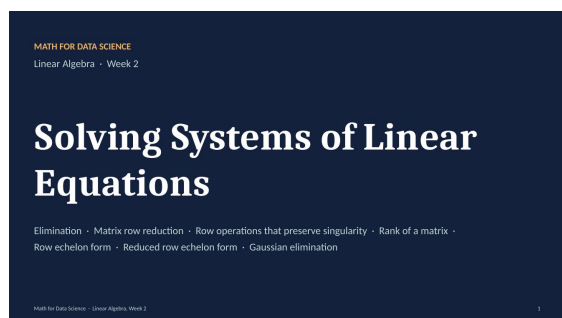


Lecture Notes: Solving Systems of Linear Equations

One entry per slide of the Week 2 deck. Each entry shows the slide, what to say, and the mathematics worked out in full (every row operation and every arithmetic step), so the notes can be used to teach from, or handed to students as a study guide. The deck has six parts: solving systems, matrix row reduction, rank, row echelon form, reduced row echelon form, and Gaussian elimination, with a machine-learning motivation up front and three in-class quizzes.



Slide 1 · Cover

Welcome back. Week 1 introduced matrices, singular and non-singular systems, and the determinant. This week we learn to actually **solve** systems of linear equations, and to turn that solution procedure into a machine-friendly algorithm on matrices.

By the end of the session students should be able to: solve a square system by elimination; recognise redundant and contradictory systems; carry out row reduction on a matrix; define and compute rank, row echelon form and reduced row echelon form; and run Gaussian elimination on an augmented matrix.

WEEK 2

Roadmap for this week

1 Solving systems of equations Non-singular, singular, and systems with more variables.	2 Matrix row reduction Turning a system into a matrix and reducing its rows.
3 Row operations that preserve singularity Swap, scale, add, and what happens to the determinant.	4 Rank of a matrix How much information a system really carries.
5 Row echelon form and reduced row echelon form Pivots, the staircase, and counting the rank.	6 Gaussian elimination The complete algorithm on the augmented matrix.

We start with the machine-learning motivation. Three short in-class quizzes follow: after non-singular systems, after singular systems, and after rank.

Math for Data Science · Linear Algebra, Week 2

Slide 2 · Roadmap

Walk through the six topics in order. The first half (topics 1 to 3) is procedural: how to solve a system and why the moves are legal. The second half (topics 4 to 6) is conceptual: rank measures information, echelon forms are the standardised end product of elimination, and Gaussian elimination packages everything into one algorithm.

Mention the three quizzes so students know pauses are coming. Each quiz is two or three minutes of individual work followed by a solution slide.

MOTIVATION

MOTIVATION

Why systems of linear equations matter in machine learning

Behind every neural network, regression model and recommender system sits a matrix. This week is about what those matrices tell us and how to solve for them.

Math for Data Science · Linear Algebra, Week 2

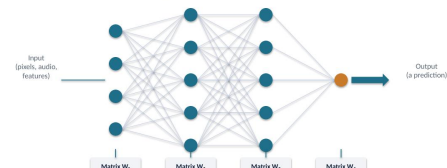
Slide 3 · Motivation divider

Before the mathematics, spend two or three minutes on why a data scientist cares. The one-sentence version: every model you will fit stores its knowledge in matrices, and fitting the model means solving for the entries of those matrices.

MACHINE LEARNING MOTIVATION

Neural networks are matrix operations

A neural network is a chain of layers. Each layer multiplies its input by a matrix of weights and adds a bias. Running the network is matrix multiplication; training it is solving for the entries of those matrices.



Everything this week (solving systems, row reduction, rank) is the toolkit for reasoning about these matrices.

Math for Data Science · Linear Algebra, Week 2

Slide 4 · Neural networks are matrix operations

A neural network takes an input vector (pixel values, audio samples, tabular features), multiplies it by a weight matrix W_1 , adds a bias and applies a simple non-linearity, then repeats with W_2 , W_3 , and so on until a prediction comes out.

Two take-aways for the class. First, **using** a trained network is nothing but repeated matrix multiplication, so matrices are the data structure of machine learning. Second, **training** a network means finding the entries of these matrices, which is a very large system of equations solved approximately. The tools of this week (solving systems, row reduction, rank) are the exact versions of the ideas that training algorithms approximate.

MACHINE LEARNING MOTIVATION

Where this shows up in data science

ACOUSTIC MONITORING

Sound recognition

Underwater microphones record the sounds of fish and marine mammals; a neural network classifies the species. Tracking species through sound helps preserve bio-habitats. The audio becomes a matrix of features, and the network is a stack of matrix operations.

GENERATIVE MODELS

AI-generated music

Automatic music generation compresses music into discrete codes, then trains a model on a specific genre to produce new pieces. Learning the compression and the generator means fitting very large matrices.

THE SIMPLEST MODEL

Linear regression

Fitting a line (or a hyperplane) to data means solving the normal equations $X^T X \beta = X^T y$; a system of linear equations in the coefficients β . Whether it has one solution, many, or none is exactly today's question.

Same mathematics in each case: data in a matrix, unknowns in a vector, equations linking them.

Math for Data Science · Linear Algebra, Week 2

Slide 5 · Where this shows up in data science

Three concrete applications. **Sound recognition:** hydrophones record underwater sound; a network classifies which species is present, which lets ecologists monitor habitats without disturbing them. **Music generation:** a model first compresses audio into discrete codes, then learns the patterns of a genre well enough to produce new pieces. Both are neural networks, so both are stacks of matrices.

Linear regression is the simplest model and the most direct link to today. Minimising squared error leads to the normal equations $X^T X \beta = X^T y$, a system of linear equations in the unknown coefficients β . Whether that system has one solution, infinitely many, or none is exactly the question we answer this week.

PART 1 · SOLVING SYSTEMS OF LINEAR EQUATIONS

PART 1

Solving systems of linear equations

From two equations in two unknowns to three equations in three unknowns, using only three legal moves.

Math for Data Science · Linear Algebra, Week 2

Slide 6 · Part 1 divider

Part 1 works entirely with equations, no matrices yet. The goal is to make the elimination procedure feel natural before we encode it.

SOLVING SYSTEMS OF EQUATIONS

Linear systems

DEFINITION

A **linear equation** in the variables $a, b, c, \dots$ is an equation of the form

$$\alpha_a a + \alpha_b b + \alpha_c c + \dots = \beta,$$

where the coefficients $\alpha_a, \alpha_b, \dots$ and the constant β are given numbers. No products or powers of the variables appear.

A **system of linear equations** is a finite collection of linear equations in the same variables, considered together.

A **solution** of the system is an assignment of numbers to the variables that satisfies every equation at the same time.

EXAMPLES

$$\begin{aligned} a + b &= 10 \\ a + 2b &= 12 \end{aligned}$$

two equations, two unknowns; the solution is $a = 8, b = 2$

$$\begin{aligned} a + b + 2c &= 12 \\ 3a - 3b - c &= 3 \end{aligned}$$

$$2a - b + 6c = 24$$

three equations, three unknowns; we solve it later today

DATA SCIENCE CONNECTION

Fitting a linear regression means solving the normal equations $X^T X \beta = X^T y$, a system of exactly this kind.

Math for Data Science · Linear Algebra, Week 2

Slide 7 · Linear systems: definition

Read the definition slowly. A **linear equation** is a weighted sum of the variables set equal to a constant; the key restriction is that variables appear only to the first power and never multiplied together. So $3a - b + 2c = 7$ is linear while $ab = 1$ or $a^2 = 4$ is not.

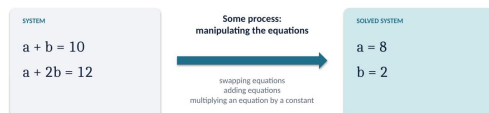
A **system** is several such equations about the same unknowns, and a **solution** must satisfy all of them simultaneously. Point out the two examples on the right: the 2×2 system whose solution is $(8, 2)$, and the 3×3 system we will solve later.

Data science connection: the normal equations of least squares are a linear system of exactly this form, with the regression coefficients as the unknowns.

SOLVING SYSTEMS OF EQUATIONS

A system and its solution

An apple and a banana cost \$10. An apple and two bananas cost \$12. What does each cost?



Reading it off: the second purchase differs from the first by exactly one banana, so a banana costs $12 - 10 = 2$, and the apple costs $10 - 2 = 8$. Elimination turns this intuition into a procedure.

Math for Data Science · Linear Algebra, Week 2

Slide 8 · A system and its solution

Use the word problem first: an apple and a banana cost \$10, an apple and two bananas cost \$12. Most students will see immediately that the extra banana costs \$2, hence the apple costs \$8. The system is $a + b = 10$, $a + 2b = 12$ and the solved system is $a = 8$, $b = 2$.

The slide names the three kinds of manipulation we are allowed: swapping equations, adding equations, and multiplying an equation by a constant. The whole of elimination is a disciplined way of applying these three moves.

Equivalent systems and the three legal operations

DEFINITION

Two systems are **equivalent** if they have exactly the same solutions. Each of the following operations replaces a system by an equivalent one:

(1) swap two equations; (2) multiply an equation by a non-zero constant; (3) add a multiple of one equation to another.

Solving a system means applying these operations until each equation involves a single variable.

MULTIPLYING BY A CONSTANT

$$\begin{array}{r} a + b = 10 \\ \times 7 \\ \hline 7a + 7b = 70 \end{array}$$

ADDING TWO EQUATIONS

$$\begin{array}{r} a + b = 10 \\ + (2a + 3b = 22) \\ \hline 3a + 4b = 32 \end{array}$$

Math for Data Science · Linear Algebra, Week 2

Slide 9 · Equivalent systems and the legal operations

Two systems are **equivalent** when they have the same solution set. The three operations (swap, scale by a non-zero constant, add a multiple of one equation to another) are legal precisely because each is reversible, so no solution is gained or lost.

Work the two mini examples: multiplying $a + b = 10$ by 7 gives $7a + 7b = 70$, which is the same line; adding $a + b = 10$ and $2a + 3b = 22$ gives $3a + 4b = 32$, which every solution of the original pair also satisfies. Ask why scaling by zero is forbidden: it would erase the equation and could not be undone.

Worked example: two equations, two unknowns

1 System

$$\begin{array}{r} 5a + b = 17 \\ 4a - 3b = 6 \end{array}$$

2 Divide by the coefficient of a

$$\begin{array}{r} a + 0.2b = 3.4 \\ a - 0.75b = 1.5 \end{array}$$

Both equations now start with a.

3 Subtract equation 1 from equation 2

$$\begin{array}{r} -0.95b = -1.9 \\ b = 2 \end{array}$$

a is eliminated from equation 2.

4 Back-substitute $b = 2$

$$\begin{array}{r} a + 0.2(2) = 3.4 \\ a = 3 \end{array}$$

Solution: $a = 3, b = 2$.

The pattern: make the leading coefficients equal, subtract to eliminate a variable, solve the smaller system, then substitute back. Every step is one of the three legal operations, so the final system is equivalent to the original.

Math for Data Science · Linear Algebra, Week 2

Slide 10 · Worked example: 2×2

System: $5a + b = 17$ and $4a - 3b = 6$. Step 1, divide each equation by its coefficient of a: $a + 0.2b = 3.4$ and $a - 0.75b = 1.5$. Step 2, subtract the first from the second: $-0.95b = -1.9$, so $b = 2$. Step 3, back-substitute: $a + 0.2(2) = 3.4$ gives $a = 3$.

Emphasise the pattern, because it is the whole algorithm: make the leading coefficients match, subtract to eliminate a variable, solve the smaller system, substitute back. Check: $5(3) + 2 = 17$ and $4(3) - 3(2) = 6$.

What if a coefficient of a is already zero?

$$\begin{array}{r} \text{SYSTEM} \\ 5a + b = 17 \\ 3b = 6 \end{array}$$

$$\begin{array}{r} \text{ISOLATE } b \\ 3b = 6 \\ b = 2 \end{array}$$

$$\begin{array}{r} \text{SOLVED} \\ a = 3 \\ b = 2 \end{array}$$

There is nothing to eliminate: a already has coefficient 0 in the second equation, so that equation gives b directly.

Then $a + 0.2(2) = 3.4$ gives $a + 0.4 = 3.4$, so $a = 3$.

In matrix language this zero is a head start: the row is already in the shape elimination is trying to produce.

Math for Data Science · Linear Algebra, Week 2

Slide 11 · A coefficient that is already zero

System: $5a + b = 17$ and $3b = 6$. There is no a to eliminate from the second equation because its coefficient is already 0. So $b = 2$ immediately, then $a + 0.2(2) = 3.4$ gives $a = 3$.

The lesson: zeros in the right places are the *goal* of elimination. When we move to matrices, a zero below the diagonal means one less step of work.

QUIZ 1 • PAUSE AND SOLVE

Solve the following system of equations

$$\begin{aligned} 2a + 5b &= 46 \\ 8a + b &= 32 \end{aligned}$$

Use the three legal operations: scale an equation, subtract to eliminate one variable, then back-substitute.

Write your answer as a pair (a, b).

Math for Data Science · Linear Algebra, Week 2

Slide 12 • Quiz 1

Give two to three minutes. System: $2a + 5b = 46$ and $8a + b = 32$. Encourage students to pick whichever variable is easier to eliminate; b is convenient here because its coefficient in the second equation is 1.

SOLVING SYSTEMS OF EQUATIONS

Quiz 1 • Solution

SYSTEM

$$\begin{aligned} 2a + 5b &= 46 \\ 8a + b &= 32 \end{aligned}$$

SOLUTION

$$a = 3, b = 8$$

ELIMINATE b

Multiply equation 2 by 5: $40a + 5b = 160$
 Subtract equation 1: $38a = 114$
a = 3

BACK-SUBSTITUTE

$$8(3) + b = 32$$
b = 8

CHECK

$$\begin{aligned} 2(3) + 5(8) &= 6 + 40 = 46 \quad \checkmark \\ 8(3) + 8 &= 24 + 8 = 32 \quad \checkmark \end{aligned}$$

Math for Data Science · Linear Algebra, Week 2

Slide 13 • Quiz 1 solution

Multiply the second equation by 5: $40a + 5b = 160$. Subtract the first: $38a = 114$, so $a = 3$. Back-substitute: $8(3) + b = 32$ gives $b = 8$. Check both equations: $2(3) + 5(8) = 46$ and $8(3) + 8 = 32$.

Students who substituted $b = 32 - 8a$ into the first equation reach the same answer. Point out that substitution and elimination are the same operations in a different order.

SOLVING SYSTEMS OF EQUATIONS

What if the system is singular? (redundant)

SYSTEM

$$\begin{aligned} a + b &= 10 \\ 2a + 2b &= 20 \end{aligned}$$

$\xrightarrow{\text{divide eq. 2 by 2}}$

$$\begin{aligned} a + b &= 10 \\ a + b &= 10 \end{aligned}$$

$\xrightarrow{\text{subtract eq. 1}}$

$$\begin{aligned} a + b &= 10 \\ 0 &= 0 \end{aligned}$$

The second equation carried no new information. After elimination only one equation survives, and one equation cannot pin down two unknowns. Pick any value $a = x$. Then $b = 10 - x$ also works; the system has infinitely many solutions, with one degree of freedom.

SOLUTION SET

$$\begin{aligned} a &= x \\ b &= 10 - x \end{aligned}$$

for every real number x

Math for Data Science · Linear Algebra, Week 2

Slide 14 • Singular system: redundant

System: $a + b = 10$ and $2a + 2b = 20$. Dividing the second by 2 gives $a + b = 10$ twice; subtracting yields $0 = 0$. The second equation carried no new information, so only one constraint remains for two unknowns.

Describe the solutions: choose any $a = x$, then $b = 10 - x$. One free parameter, so infinitely many solutions. Connect to Week 1: this is the singular case with determinant $1 \cdot 2 - 1 \cdot 2 = 0$.

SOLVING SYSTEMS OF EQUATIONS

What if the system is singular? (contradictory)

SYSTEM

$$\begin{aligned} a + b &= 10 \\ 2a + 2b &= 24 \end{aligned}$$

$\xrightarrow{\text{divide eq. 2 by 2}}$

$$\begin{aligned} a + b &= 10 \\ a + b &= 12 \end{aligned}$$

$\xrightarrow{\text{subtract eq. 1}}$

$$\begin{aligned} a + b &= 10 \\ 0 &= 2 \end{aligned}$$

A contradiction. No choice of a and b can make 0 equal to 2, so no pair satisfies both equations. The left-hand sides are proportional ($2a + 2b$ is twice $a + b$) but the right-hand sides are not (24 is not twice 10).

SOLUTION SET

empty

the system has no solutions

Math for Data Science · Linear Algebra, Week 2

Slide 15 • Singular system: contradictory

System: $a + b = 10$ and $2a + 2b = 24$. Dividing gives $a + b = 12$, and subtracting the first equation produces $0 = 2$, which is false. No pair (a, b) can satisfy both equations.

Geometrically the two equations are parallel lines. Note that the coefficient matrix is the same as on the previous slide; only the constants differ. Singularity is a property of the coefficients; whether the singular system has many or no solutions depends on the constants.

Classifying systems

DEFINITION

A system is **consistent** if it has at least one solution and **inconsistent** if it has none.

A square system (as many equations as unknowns) is **non-singular** if it has exactly one solution, and **singular** otherwise, that is, when it has infinitely many solutions or none.

Unique solution

NON-SINGULAR

$$a + b = 10$$

$$a + 2b = 12$$

Geometrically: two lines that cross in one point.

Infinitely many solutions

SINGULAR - REDUNDANT

$$a + b = 10$$

$$2a + 2b = 20$$

Geometrically: two equations for the same line.

No solution

SINGULAR - CONTRADICTORY

$$a + b = 10$$

$$2a + 2b = 24$$

Geometrically: two parallel lines

Slide 16 · Classifying systems

Collect the vocabulary. **Consistent**: at least one solution. **Inconsistent**: none. For square systems, **non-singular** means exactly one solution and **singular** covers both the redundant case (infinitely many) and the contradictory case (none).

The three cards are the three pictures to keep in mind: two lines crossing at a point; two descriptions of the same line; two parallel lines.

Solve the following system of equations

$$5a + b = 11$$

$$10a + 2b = 22$$

Look closely at the two equations before you start computing.

Is the system non-singular, redundant, or contradictory? Describe every solution.

Slide 17 · Quiz 2

System: $5a + b = 11$ and $10a + 2b = 22$. Ask students to look before computing. Two minutes.

Quiz 2 · Solution

SYSTEM

$$5a + b = 11$$

$$10a + 2b = 22$$

ELIMINATE

Divide equation 2 by 2: $5a + b = 11$

This is equation 1 again. Subtracting gives

$$0 = 0$$

The system is singular and redundant: equation 2 is just twice equation 1, so it adds no information.

One equation in two unknowns leaves one degree of freedom, so there are infinitely many solutions.

CHECK
 $\{a, b\} = \{(0, 11), (1, 6), (2, 1), (-1, 16), \dots\}$ all satisfy both equations.

Slide 18 · Quiz 2 solution

The second equation is exactly twice the first, so after dividing by 2 and subtracting we get $0 = 0$. The system is singular and redundant: infinitely many solutions, described by $b = 11 - 5a$ for any real a . Give examples: $(0, 11)$, $(1, 6)$, $(2, 1)$.

If someone computed the determinant, $5 \cdot 2 - 1 \cdot 10 = 0$ confirms singularity, but the determinant alone cannot distinguish redundant from contradictory; for that we need the constants.

Elimination with three variables (1 of 3)

SYSTEM

$$a + b + 2c = 12$$

$$3a - 3b - c = 3$$

$$2a - b + 6c = 24$$

Three equations, three unknowns.

DIVIDE EACH ROW BY ITS COEFFICIENT OF a

$$a + b + 2c = 12$$

$$a - b - \frac{1}{3}c = 1$$

$$a - \frac{1}{2}b + 3c = 12$$

Now every equation starts with a .

SUBTRACT EQUATION 1 FROM THE OTHERS

$$a + b + 2c = 12$$

$$-2b - \frac{7}{3}c = -11$$

$$-3/2 b + c = 0$$

a is isolated in equation 1; equations 2 and 3 form a 2×2 system in b and c .

Same idea as before: use the first equation to remove a from every other equation, then solve what is left.

Slide 19 · Three variables, step 1

System: $a + b + 2c = 12$, $3a - 3b - c = 3$, $2a - b + 6c = 24$. Divide each equation by its coefficient of a : $a + b + 2c = 12$, $a - b - \frac{1}{3}c = 1$, $a - \frac{1}{2}b + 3c = 12$. Subtract the first from the other two: $-2b - \frac{7}{3}c = -11$ and $-3/2 b + c = 0$.

The variable a now appears only in the first equation. What remains below it is a 2×2 system in b and c , which we already know how to solve. This is the recursive heart of elimination.

Elimination with three variables (2 of 3)

THE SYSTEM SO FAR

$$\begin{aligned} a + b + 2c &= 12 \\ -2b - 7/3c &= -11 \\ -3/2b + c &= 0 \end{aligned}$$

Work on the last two equations only.

DIVIDE BY THE COEFFICIENT OF b

$$\begin{aligned} a + b + 2c &= 12 \\ b + 7/6c &= 11/2 \\ b - 2/3c &= 0 \end{aligned}$$

Both now start with b.

SUBTRACT EQUATION 2 FROM EQUATION 3

$$\begin{aligned} a + b + 2c &= 12 \\ b + 7/6c &= 11/2 \\ -11/6c &= -11/2 \end{aligned}$$

b is isolated in equation 2, and the last equation gives $c = 3$.

The system is now triangular: the first equation has a, b, c; the second has b, c; the third has c alone.

Math for Data Science · Linear Algebra, Week 2

Slide 20 · Three variables, step 2

Divide the last two equations by their coefficient of b: $b + 7/6c = 11/2$ and $b - 2/3c = 0$. Subtract the second from the third: $(-2/3 - 7/6)c = -11/2$, that is $-11/6c = -11/2$, so $c = 3$.

The system is now triangular: the first equation involves a, b, c; the second b, c; the third c alone. Triangular systems are solved by reading from the bottom up.

Three variables (3 of 3): back substitution

TRIANGULAR SYSTEM

$$\begin{aligned} a + b + 2c &= 12 \\ b + 7/6c &= 11/2 \\ c &= 3 \end{aligned}$$

Back substitution: solve the last equation first and feed each answer into the equation above it.

SUBSTITUTE UPWARDS

 $c = 3$ into equation 2: $b + 7/6 \cdot 3 = 11/2$, so $b + 7/2 = 11/2$, so $b = 2$ $b = 2, c = 3$ into equation 1: $a + 2 + 6 = 12$, so $a = 4$ **Solution: $a = 4, b = 2, c = 3$**

CHECK
 $4 + 2 + 2(3) = 12 \checkmark$
 $2(4) - 3(2) = 3 = 3 \checkmark$
 $2(4) - 2 + 6(3) = 24 \checkmark$

Math for Data Science · Linear Algebra, Week 2

Slide 21 · Three variables, step 3: back substitution

From $c = 3$, the second equation gives $b + 7/6 \cdot 3 = 11/2$, that is $b + 7/2 = 11/2$, so $b = 2$. The first gives $a + 2 + 6 = 12$, so $a = 4$. Solution $(a, b, c) = (4, 2, 3)$.

Check all three original equations: $4 + 2 + 6 = 12$; $12 - 6 - 3 = 3$; $8 - 2 + 18 = 24$. Checking is cheap and catches arithmetic slips.

PART 2 · MATRIX ROW REDUCTION

PART 2

Matrix row reduction

The same elimination steps, written on the coefficients only. Rows replace equations.

Math for Data Science · Linear Algebra, Week 2

Slide 22 · Part 2 divider

We now strip away the variable names and work on the numbers alone. Nothing new happens mathematically; the notation becomes compact enough for a computer.

From systems to matrices

DEFINITION

The **coefficient matrix** of a system lists the coefficients of the variables: one row per equation, one column per variable. **Row reduction** applies the three legal operations to the rows of the matrix instead of to the equations.

ORIGINAL SYSTEM

$$\begin{aligned} 5a + b &= 17 \\ 4a - 3b &= 6 \end{aligned}$$

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix}$$

Original matrix

INTERMEDIATE SYSTEM

$$\begin{aligned} a + 0.2b &= 3.4 \\ b &= 2 \end{aligned}$$

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Upper triangular: row echelon form

SOLVED SYSTEM

$$\begin{aligned} 1a + 0b &= 3 \\ 0a + 1b &= 2 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Diagonal: reduced row echelon form

Math for Data Science · Linear Algebra, Week 2

Slide 23 · From systems to matrices

The **coefficient matrix** has one row per equation and one column per variable. For $5a + b = 17$, $4a - 3b = 6$ the matrix is $\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix}$. The elimination steps from the earlier slide turn it into $\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$ (upper triangular, called **row echelon form**) and finally into the identity $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (**reduced row echelon form**).

Read the three matrices against the three systems above them so students see that the matrix is just bookkeeping for the equations. The constants are left out here; they return with the augmented matrix in Part 6.

Singular systems as matrices

ORIGINAL SYSTEM

$$\begin{aligned} a + b &= 10 \\ 2a + 2b &= 20 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$



$$\begin{aligned} a + b &= 10 \\ 0a + 0b &= 0 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

redundant

$$\begin{aligned} 5a + b &= 11 \\ 10a + 2b &= 22 \end{aligned}$$

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix}$$



$$\begin{aligned} a + 0.2b &= 2.2 \\ 0a + 0b &= 0 \end{aligned}$$

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$$

redundant

$$\begin{aligned} 0a + 0b &= 1 \\ 0a + 0b &= 0 \end{aligned}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$



$$\begin{aligned} 0a + 0b &= 0 \\ 0a + 0b &= 0 \end{aligned}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

no information at all

A row of zeros appears in the row echelon form exactly when the system is singular.

Math for Data Science - Linear Algebra, Week 2

Slide 24 • Singular systems as matrices

The redundant system $a + b = 10$, $2a + 2b = 20$ has matrix $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$; elimination gives $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. The quiz system $5a + b = 11$, $10a + 2b = 22$ gives $\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix}$ and then $\begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$. The all-zero system stays all zero.

Highlight the rule: a singular matrix produces a row of zeros during elimination. That row is the matrix version of the equation $0 = 0$ (or $0 = \text{something}$, once constants are attached).

Elementary row operations

DEFINITION

The three **elementary row operations** on a matrix are: swapping two rows, multiplying a row by a non-zero scalar, and adding a multiple of one row to another row.

Theorem. Elementary row operations do not change the solution set of the corresponding system, and they do not change whether the matrix is singular.

Swap

$$R_i \leftrightarrow R_j$$

Reorders the equations. Nothing about the solutions changes.

reversible

Scale

$$R_i \leftarrow k \cdot R_i, k \neq 0$$

Multiplies one equation by a constant. Dividing by k undoes it.

reversible

Replace

$$R_i \leftarrow R_i + k \cdot R_j$$

Adds a multiple of another equation. Subtracting $k \cdot R_j$ undoes it.

reversible

Math for Data Science - Linear Algebra, Week 2

Slide 25 • Elementary row operations

Define the three **elementary row operations**: swap $R_i \leftrightarrow R_j$; scale $R_i \leftarrow k \cdot R_i$ with $k \neq 0$; replace $R_i \leftarrow R_i + k \cdot R_j$. They are the three legal moves on equations, rewritten for rows.

State the theorem: these operations do not change the solution set and do not change singularity. The reason is reversibility; each operation has an inverse of the same type (swap again, scale by $1/k$, subtract $k \cdot R_j$).

Row operations and the determinant

Start from the same matrix each time and watch what happens to $\det = 5 \cdot 3 - 1 \cdot 4 = 11$.

Switch the rows

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 3 \\ 5 & 1 \end{bmatrix}$$

$$\det = 11$$

$$\det = 4 \cdot 1 - 3 \cdot 5 = -11$$

The sign flips.

Multiply row 1 by 10

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 50 & 10 \\ 4 & 3 \end{bmatrix}$$

$$\det = 11$$

$$\det = 50 \cdot 3 - 10 \cdot 4 = 110 = 10 \cdot 11$$

The determinant is multiplied by 10.

Add row 1 to row 2

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 5 & 1 \\ 9 & 4 \end{bmatrix}$$

$$\det = 11$$

$$\det = 5 \cdot 4 - 1 \cdot 9 = 11$$

The determinant does not change.

A determinant that is zero stays zero, and one that is non-zero stays non-zero: row operations preserve singularity.

Math for Data Science - Linear Algebra, Week 2

Slide 26 • Row operations and the determinant

Start from $\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix}$ with determinant $15 - 4 = 11$. Swapping the rows gives $\begin{bmatrix} 4 & 3 \\ 5 & 1 \end{bmatrix}$ with determinant $4 - 15 = -11$: the sign flips. Multiplying row 1 by 10 gives $\begin{bmatrix} 50 & 10 \\ 4 & 3 \end{bmatrix}$ with determinant $150 - 40 = 110 = 10 \cdot 11$. Adding row 1 to row 2 gives $\begin{bmatrix} 5 & 1 \\ 9 & 4 \end{bmatrix}$ with determinant $20 - 9 = 11$: unchanged.

The point for this week: a zero determinant stays zero and a non-zero one stays non-zero under all three operations, which is the determinant's way of saying that row operations preserve singularity.

PART 3 • RANK OF A MATRIX

PART 3

Rank of a matrix

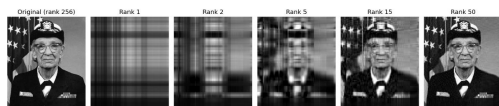
Two equations do not always carry two pieces of information. Rank measures how much they really say.

Math for Data Science - Linear Algebra, Week 2

Slide 27 • Part 3 divider

Rank is the first genuinely new concept. Motivate it with the question: when do two equations really give two pieces of information?

Compressing images: reducing rank



A grayscale image is a matrix: one number per pixel. This one is 300×256 and has rank 256, so it carries 256 independent pieces of information.

Keeping only the k most important pieces (a rank- k approximation) rebuilds the picture from far fewer numbers: rank 1 is just stripes, rank 15 is recognisable, rank 50 is close to the original.

Rank is the amount of information in a matrix. The same idea powers image compression, principal component analysis and recommender systems.

Math for Data Science · Linear Algebra, Week 2

Slide 28 · Compressing images: reducing rank

A grayscale image is a matrix of pixel intensities. This one is 300×256 and has rank 256. A rank- k approximation keeps only the k most important 'pieces of information' and rebuilds the picture from k row-patterns and k column-patterns.

Walk along the panel: rank 1 is horizontal and vertical stripes; rank 5 shows a silhouette; rank 15 is recognisable; rank 50 is close to the original using a fraction of the numbers. The method behind it (the singular value decomposition) comes later in the course, but the concept of rank as information content is what we define now. The same idea underlies principal component analysis and recommender systems.

Systems of information

System 1 The dog is black. The cat is orange. Two sentences Two pieces of information Rank = 2	System 2 The dog is black. The dog is black. Two sentences One piece of information Rank = 1	System 3 The dog. The dog. Two sentences Zero pieces of information Rank = 0
---	---	---

Rank counts independent pieces of information, not the number of statements.

Math for Data Science · Linear Algebra, Week 2

Slide 29 · Systems of information

A non-mathematical warm-up. 'The dog is black. The cat is orange.' is two sentences and two facts: rank 2. 'The dog is black' twice is two sentences but one fact: rank 1. 'The dog. The dog.' says nothing: rank 0.

Rank counts independent pieces of information, not the number of statements. Keep this slogan; every later definition of rank is a precise version of it.

Rank of a 2×2 system

System 1 $a + b = 0$ $a + 2b = 0$ $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ Two equations Two pieces of information Rank = 2	System 2 $a + b = 0$ $2a + 2b = 0$ $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ Two equations One piece of information Rank = 1	System 3 $0a + 0b = 0$ $0a + 0b = 0$ $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ Two equations Zero pieces of information Rank = 0
---	---	---

The rank of the matrix is the rank of the system it represents.

Math for Data Science · Linear Algebra, Week 2

Slide 30 · Rank of a 2×2 system

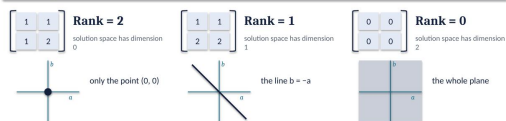
Translate the sentences into equations. $a + b = 0$ and $a + 2b = 0$ are two independent constraints: rank 2. $a + b = 0$ and $2a + 2b = 0$ say the same thing: rank 1. $0a + 0b = 0$ twice says nothing: rank 0.

The rank of the matrix is defined as the rank of the system it represents. Students should notice that the rank-1 and rank-0 matrices are exactly the singular ones from Week 1.

Rank and the solution space

DEFINITION

For a homogeneous system (all constants equal to 0) the **solution space** is the set of all its solutions; its **dimension** is the number of free parameters needed to describe it. **Rank of the matrix = number of variables - dimension of the solution space.**



With two variables: **Rank = 2 - (dimension of the solution space).**

Slide 31 • Rank and the solution space

For a homogeneous system (all constants 0) the **solution space** is the set of all solutions. Rank 2: only (0, 0) works, a single point, dimension 0. Rank 1: every point on the line $b = -a$ works, dimension 1. Rank 0: every point in the plane works, dimension 2.

The formula: rank = number of variables - dimension of the solution space, here rank = 2 - dim. This is a first glimpse of the rank-nullity theorem. Draw the three pictures on the board if students have not met 'dimension' informally.

Rank and singularity



FACT

A square $n \times n$ matrix is non-singular exactly when its rank equals n (full rank). Any smaller rank means at least one row carries no new information: the matrix is singular.

Data science connection. If two feature columns are proportional (collinear), $X^T X$ loses rank and least squares has no unique solution.

Slide 32 • Rank and singularity

Put the two ideas together: an $n \times n$ matrix is non-singular exactly when its rank is n (**full rank**). Any smaller rank means a row that carries no new information, hence a zero row after elimination, hence a zero determinant.

Data science connection: in regression, if two feature columns are proportional (perfectly collinear), $X^T X$ loses rank, becomes singular, and the least-squares coefficients are not uniquely determined. Software then warns or drops a column.

Find the rank of each matrix

MATRIX 1

$$\begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}$$

MATRIX 2

$$\begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$$

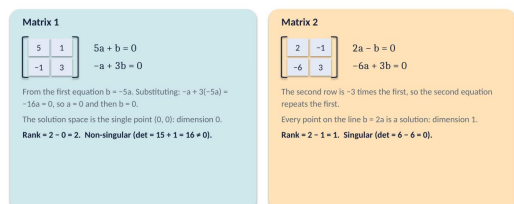
Hint: write the homogeneous system (constants 0), find its solution space, and use

Rank = 2 - dimension of the solution space.

Slide 33 • Quiz 3

Matrix 1 is $\begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}$; Matrix 2 is $\begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$. Ask for the rank of each, using the solution space of the homogeneous system or any other method. Three minutes.

Quiz 3 • Solution



Slide 34 • Quiz 3 solution

Matrix 1: the system $5a + b = 0$, $-a + 3b = 0$ gives $b = -5a$, then $-a - 15a = -16a = 0$, so $a = b = 0$. Solution space is a point, dimension 0, rank $2 - 0 = 2$. Non-singular, and indeed $\det = 15 + 1 = 16$.

Matrix 2: the second row is -3 times the first, so both equations say $2a - b = 0$. Solution space is the line $b = 2a$, dimension 1, rank 1. Singular, $\det = 6 - 6 = 0$.

Rank of a 3×3 system

Math for Data Science - Linear Algebra, Week 2

Four systems with three equations each. System 1 ($a + b + c, a + 2b + c, a + b + 2c$, all $= 0$) has three independent equations: rank 3. System 2 ($a + b + c, a + b + 2c, a + b + 3c$) has only two independent ones, since the third is $2 \cdot (\text{second}) - (\text{first})$: rank 2. System 3 (rows proportional to 1, 1, 1) has one: rank 1. The zero system has rank 0.

Counting independent equations by eye is already awkward at 3×3 . That is the cue for the next slide.

An easier way to compute the rank?

Yes. Bring the matrix to row echelon form and count the pivots. Equivalently, count the ones on the diagonal of its reduced row echelon form. The next part makes both forms precise.

Answer: bring the matrix to row echelon form and count the pivots. Equivalently, count the ones on the diagonal of the reduced row echelon form. The next two parts define these forms precisely and show that elimination always reaches them.

PART 4

A staircase shape that elimination always reaches, and the fastest way to read off the rank

Row echelon form is the standard shape that elimination produces. Once students can recognise it, rank becomes a counting exercise.

Row echelon form: definition

A matrix is in row echelon form if

- (1) all rows consisting entirely of zeros are at the bottom;
 - (2) in each non-zero row the first non-zero entry, called the **pivot**, lies strictly to the right of the pivot of the row above.
- The **rank** of a matrix is the number of pivots in its row echelon form.

Note. In general pivots other than 1 are allowed. In this class we divide each row by its pivot so that every pivot is 1: this makes no mathematical difference.



A matrix is in **row echelon form** if all zero rows are at the bottom and, in every non-zero row, the first non-zero entry (the **pivot**) sits strictly to the right of the pivot in the row above. The pivots therefore form a staircase descending to the right. The **rank** is the number of pivots.

Point at the two 5×5 patterns: pivots 2, 1, 3, -5 , 1 on the diagonal give rank 5; pivots 3, 1, -4 followed by two zero rows give rank 3. Note: pivots need not equal 1 in general; in this class we normalise them to 1 for tidiness, and this changes nothing.

Reaching row echelon form, step by step



THE STEPS

Step 1 divide each row by its leftmost coefficient: $5a + b = 17$ becomes $a + 0.2b = 3.4$, $4a - 3b = 6$ becomes $a - 0.75b = 1.5$

Step 2 subtract row 1 from row 2: $[1 \ -0.75] - [1 \ 0.2] = [0 \ -0.95]$

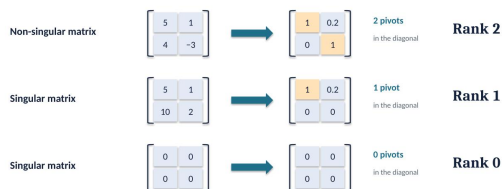
Step 3 divide row 2 by its leftmost non-zero coefficient: $[0 \ -0.95] \div (-0.95) = [0 \ 1]$

Slide 39 • Reaching row echelon form, step by step

Matrix $[5 \ 1; 4 \ -3]$. Step 1: divide each row by its leftmost coefficient to get $[1 \ 0.2; 1 \ -0.75]$. Step 2: subtract row 1 from row 2 to get $[1 \ 0.2; 0 \ -0.95]$. Step 3: divide row 2 by -0.95 to get $[1 \ 0.2; 0 \ 1]$, which is in row echelon form with two pivots.

These are the same three moves as the 2×2 worked example in Part 1, now written on rows. Students should be able to say which elementary row operation each step is.

Row echelon form, singularity and rank



Full rank (2 pivots) means non-singular; a missing pivot means a zero row and a singular matrix.

Slide 40 • Row echelon form, singularity and rank

Three matrices and their echelon forms. $[5 \ 1; 4 \ -3] \rightarrow [1 \ 0.2; 0 \ 1]$: two pivots, rank 2, non-singular. $[5 \ 1; 10 \ 2] \rightarrow [1 \ 0.2; 0 \ 0]$: one pivot, rank 1, singular. $[0 \ 0; 0 \ 0]$ stays zero: no pivots, rank 0, singular.

Full rank means a pivot in every row; a missing pivot means a zero row, which is the fingerprint of a singular matrix.

Row echelon form of a 3×3 system

SYSTEM

$$a + b + 2c = 12$$

$$3a - 3b - c = 3$$

$$2a - b + 6c = 24$$



$$R_2 \leftarrow R_2 - 3R_1, \quad R_3 \leftarrow R_3 - 2R_1$$

$$\text{then } R_3 \leftarrow R_3 - R_2$$

Three pivots: rank 3. The last row gives $c = 3$ at once, exactly as in the elimination we did by hand.

TRIANGULAR SYSTEM

$$a + b + 2c = 12$$

$$-6b - 7c = -33$$

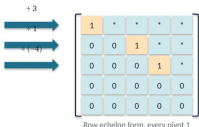
$$11c = 33$$

Slide 41 • Row echelon form of a 3×3 system

The 3×3 system from Part 1, now as the matrix $[1 \ 1 \ 2; 3 \ -3 \ -1; 2 \ -1 \ 6]$. Operations: $R_2 \leftarrow R_2 - 3R_1$ gives $[0 \ -6 \ -7]$; $R_3 \leftarrow R_3 - 2R_1$ gives $[0 \ -3 \ 2]$; then $R_3 \leftarrow 2R_3 - R_2$ gives $[0 \ 0 \ 11]$. Echelon form $[1 \ 1 \ 2; 0 \ -6 \ -7; 0 \ 0 \ 11]$, three pivots, rank 3.

Carrying the constants along (12, 3, 24) gives 12, -33 , 33 and the triangular system $a + b + 2c = 12$, $-6b - 7c = -33$, $11c = 33$, so $c = 3$ as before. The last operation is two legal moves in one (scale R_3 by 2, then subtract R_2); it avoids fractions.

Normalizing the pivots to 1



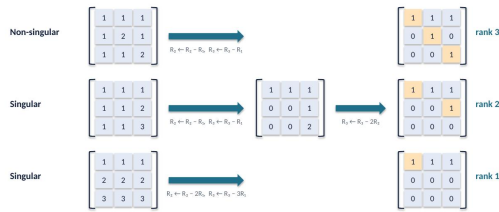
WHY IT IS ALLOWED

Dividing a row by its pivot is a legal row operation (scaling by a non-zero constant).
The staircase, the number of pivots and the rank are unchanged.

Slide 42 • Normalizing the pivots

Dividing each non-zero row by its pivot (here $\div 3$, $\div 1$, $\div (-4)$) is a scaling operation, so it is legal. The staircase, the number of pivots and the rank are unchanged; only the convention changes. We adopt pivots equal to 1 because it makes the next step, reduced row echelon form, cleaner.

More examples



The second example needs two rounds: after the first round, row 2 is twice row 1.
Math for Data Science - Linear Algebra, Week 2

Slide 43 • More examples

[1 1 1; 1 2 1; 1 1 2]: subtract row 1 from rows 2 and 3 to get [1 1 1; 0 1 0; 0 0 1], rank 3, non-singular. [1 1 1; 1 1 2; 1 1 3]: the same step gives [1 1 1; 0 0 1; 0 0 2]; then $R_3 \leftarrow R_3 - 2R_2$ gives a zero row, so rank 2. [1 1 1; 2 2 2; 3 3 3]: subtracting $2R_1$ and $3R_1$ leaves two zero rows, rank 1.

The middle example shows that one round of elimination may not be enough; keep going until the staircase condition holds in every row.

Pivots count the rank

MATRIX

Matrix 1

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Matrix 2

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 3 \end{bmatrix}$$

Matrix 3

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$$

Matrix 4

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

ROW ECHELON FORM

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Number of pivots = 3

Rank = 3

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Number of pivots = 2

Rank = 2

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Number of pivots = 1

Rank = 1

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Number of pivots = 0

Rank = 0

Math for Data Science - Linear Algebra, Week 2

43

Slide 44 • Pivots count the rank

The four 3x3 matrices from the rank slide, each with its row echelon form and highlighted pivots: 3, 2, 1 and 0 pivots, matching the ranks 3, 2, 1, 0 we found by counting independent equations. This closes the loop: rank by information content and rank by pivot count agree.

PART 5 • REDUCED ROW ECHELON FORM

PART 5

Reduced row echelon form

Continue past the staircase: pivots become 1 and everything above a pivot becomes 0.

Math for Data Science - Linear Algebra, Week 2

Slide 45 • Part 5 divider

Reduced row echelon form continues elimination upwards. It is the form in which the solution can be read off directly.

REDUCED ROW ECHELON FORM

Reduced row echelon form: definition

DEFINITION

A matrix is in **reduced row echelon form** if

- (1) it is in row echelon form;
- (2) every pivot equals 1;
- (3) every pivot is the only non-zero entry in its column, so all entries above a pivot are 0.

Fact. Every matrix has exactly one reduced row echelon form; the rank is still the number of pivots.

For a non-singular square matrix the reduced row echelon form is the identity matrix, and the augmented system is then solved outright.

Math for Data Science - Linear Algebra, Week 2

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{Rank 5}$$

- Is in row echelon form.
- Each pivot is a 1.
- Any number above a pivot is 0.
- Rank of the matrix is the number of pivots.

Slide 46 • Reduced row echelon form: definition

A matrix is in **reduced row echelon form** if it is in row echelon form, every pivot equals 1, and every pivot is the only non-zero entry in its column (zeros above as well as below). Unlike row echelon form, this form is **unique** for each matrix.

For a non-singular square matrix the reduced row echelon form is the identity. In the 5x5 rank-3 pattern, the non-pivot columns keep their entries; those columns correspond to free variables of the system.

Reduced row echelon form: a 2x2 example

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Row echelon form

$$R_1 \leftarrow R_1 - 0.2R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Reduced row echelon form

The only entry above a pivot was the 0.2. One row operation clears it.

THE COMPUTATION

$$0.2 \times \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0.2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0.2 \end{bmatrix} - \begin{bmatrix} 0 & 0.2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad \text{the entry above the second pivot is now 0}$$

Slide 47 · Reduced row echelon form: 2x2

From $\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$, the only entry above a pivot is 0.2.

Apply $R_1 \leftarrow R_1 - 0.2R_2$: $0.2 \times \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0.2 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0.2 \end{bmatrix} - \begin{bmatrix} 0 & 0.2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}$. Result $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the identity.

In the language of equations, this is substituting $b = 2$ back into $a + 0.2b = 3.4$ to isolate a . Back substitution and 'clearing above the pivots' are the same thing.

Reduced row echelon form: the general recipe

$$\begin{bmatrix} 3 & * & * & * & * \\ 0 & 0 & 2 & * & * \\ 0 & 0 & 0 & -4 & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Row echelon form

divide each row by its pivot

$$\begin{bmatrix} 1 & * & * & * & * \\ 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

clear entries above pivots

$$\begin{bmatrix} 1 & 0 & 0 & 0 & * \\ 0 & 0 & 1 & 0 & * \\ 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Reduced row echelon form

Columns without a pivot keep their entries (the stars): they correspond to free variables of the system. The pivot columns become columns of the identity matrix.

Slide 48 · Reduced row echelon form: general recipe

Two steps after reaching row echelon form. First divide each row by its pivot so every pivot is 1. Then, for each pivot, subtract multiples of its row from the rows above to turn every entry above the pivot into 0.

Pivot columns become columns of the identity matrix; columns without a pivot keep whatever entries remain, and each such column is a free variable. The number of pivots, and so the rank, never changes.

Reduced row echelon form: a 3x3 example

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

Row echelon form

$$R_1 \leftarrow R_1 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \leftarrow R_1 + 5R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 4R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reduced row echelon form

Work from the last pivot upwards, or from the first pivot downwards: either order clears every entry above a pivot and never disturbs a column already finished. The result is the identity, so the matrix has rank 3 and is non-singular.

Slide 49 · Reduced row echelon form: 3x3 example

Start from $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$. $R_1 \leftarrow R_1 - 2R_2$ gives $\begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$. $R_1 \leftarrow R_1 + 5R_3$ gives $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$. $R_2 \leftarrow R_2 - 4R_3$ gives $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The result is the identity, so the matrix has rank 3.

Either order works (last pivot upwards or first pivot downwards) because clearing a column never disturbs a column already finished.

PART 6 · THE GAUSSIAN ELIMINATION ALGORITHM

PART 6

The Gaussian elimination algorithm

Putting it all together on the augmented matrix: pivot, reduce, back-substitute, and read the answer.

Slide 50 · Part 6 divider

Gaussian elimination is the complete algorithm: everything from Parts 1 to 5, applied to the augmented matrix so the constants travel with the coefficients.

The augmented matrix

DEFINITION

The **augmented matrix** $[A \mid b]$ of the system $Ax = b$ is the coefficient matrix A with the column of constants b appended on the right. Row operations act on entire rows, constants included, so the system stays equivalent.

SYSTEM

$$\begin{aligned} 2a - b + c &= 1 \\ 2a + 2b + 4c &= -2 \\ 4a + b &= 4 \end{aligned}$$

Each row is one equation; the bar separates the coefficients of a, b, c from the constant. Now proceed with the elimination method on the rows.

AUGMENTED MATRIX

$$\begin{array}{ccc|c} R_1 & 2 & -1 & 1 \\ R_2 & 2 & 2 & -2 \\ R_3 & 4 & 1 & 0 \end{array}$$

Slide 51 • The augmented matrix

The **augmented matrix** $[A \mid b]$ appends the constants column to the coefficient matrix. For $2a - b + c = 1$, $2a + 2b + 4c = -2$, $4a + b = 4$ it is $[2 \ -1 \ 1 \mid 1; 2 \ 2 \ 4 \mid -2; 4 \ 1 \ 0 \mid 4]$. Note the 0 for the missing c in the third equation.

Row operations act on whole rows including the constant, which is exactly what we did to equations in Part 1. The vertical bar is just a reminder of where the equals signs were.

Pivoting on the first column

$$\begin{array}{ccc|c} 2 & -1 & 1 & 1 \\ 2 & 2 & 4 & -2 \\ 4 & 1 & 0 & 4 \end{array} \rightarrow \begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 2 & 2 & 4 & -2 \\ 4 & 1 & 0 & 4 \end{array} \rightarrow \begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 3 & 3 & -3 \\ 0 & 3 & -2 & 2 \end{array}$$

Start: pivot 2 in row 1 $R_1 \leftarrow \frac{1}{2}R_1$ $R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - 4R_1$

THE ROW ARITHMETIC

$$\begin{aligned} R_1 &\leftarrow \frac{1}{2}R_1: [2 \ 2 \ 4 \mid -2] \rightarrow 2 \cdot [1 \ -\frac{1}{2} \ \frac{1}{2} \mid \frac{1}{2}] = [0 \ 3 \ 3 \mid -3] \\ R_3 &\leftarrow R_3 - 4R_1: [4 \ 1 \ 0 \mid 4] \rightarrow 4 \cdot [1 \ -\frac{1}{2} \ \frac{1}{2} \mid \frac{1}{2}] = [0 \ 3 \ -2 \mid 2] \end{aligned}$$

Slide 52 • Pivoting on the first column

$R_1 \leftarrow \frac{1}{2}R_1$ makes the first pivot 1: $[1 \ -\frac{1}{2} \ \frac{1}{2} \mid \frac{1}{2}]$. Then $R_2 \leftarrow R_2 - 2R_1$: $[2 \ 2 \ 4 \mid -2] - [2 \ -1 \ 1 \mid 1] = [0 \ 3 \ 3 \mid -3]$. And $R_3 \leftarrow R_3 - 4R_1$: $[4 \ 1 \ 0 \mid 4] - [4 \ -2 \ 2 \mid 2] = [0 \ 3 \ -2 \mid 2]$.

Say out loud what 'pivoting on a column' means: choose the pivot, scale it to 1, and use it to zero out everything below it.

Pivoting on the second and third columns

$$\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 3 & 3 & -3 \\ 0 & 3 & -2 & 2 \end{array} \rightarrow \begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -5 & 5 \end{array} \rightarrow \begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{array}$$

$R_2 \leftarrow \frac{1}{3}R_2$ $R_3 \leftarrow R_3 - 3R_2$, then $R_1 \leftarrow R_1 - \frac{1}{2}R_2$ Row echelon form

THE ROW ARITHMETIC

$$\begin{aligned} R_1 &\leftarrow R_1 - \frac{1}{2}R_2: [1 \ -\frac{1}{2} \ \frac{1}{2} \mid \frac{1}{2}] - \frac{1}{2} \cdot [0 \ 1 \ 1 \mid -1] = [1 \ 0 \ -\frac{1}{2} \mid \frac{1}{2}] \\ R_3 &\leftarrow R_3 - 3R_2: [0 \ 3 \ -2 \mid 2] - 3 \cdot [0 \ 1 \ 1 \mid -1] = [0 \ 0 \ -5 \mid 5] \end{aligned}$$

AS EQUATIONS

$$\begin{aligned} a - \frac{1}{2}b + \frac{1}{2}c &= \frac{1}{2} \\ b + c &= -1 \\ c &= -1 \end{aligned}$$

Slide 53 • Pivoting on the second and third columns

$R_2 \leftarrow \frac{1}{3}R_2$ gives $[0 \ 1 \ 1 \mid -1]$. $R_3 \leftarrow R_3 - 3R_2$: $[0 \ 3 \ -2 \mid 2] - [0 \ 3 \ 3 \mid -3] = [0 \ 0 \ -5 \mid 5]$. $R_3 \leftarrow -\frac{1}{5}R_3$ gives $[0 \ 0 \ 1 \mid -1]$. The augmented matrix is now in row echelon form: $[1 \ -\frac{1}{2} \ \frac{1}{2} \mid \frac{1}{2}; 0 \ 1 \ 1 \mid -1; 0 \ 0 \ 1 \mid -1]$.

As equations: $a - \frac{1}{2}b + \frac{1}{2}c = \frac{1}{2}$, $b + c = -1$, $c = -1$. Three pivots, rank 3, so the system is non-singular and has a unique solution.

Back substitution

$$\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{array} \rightarrow \begin{array}{ccc|c} 1 & -\frac{1}{2} & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array} \rightarrow \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array}$$

Row echelon form $R_1 \leftarrow R_1 - \frac{1}{2}R_2$, $R_2 \leftarrow R_2 - \frac{1}{2}R_3$ $R_1 \leftarrow R_1 + \frac{1}{2}R_2$

THE RESULT

$$a = 1, b = 0, c = -1$$

The coefficient part is the identity matrix; the constants column is the solution.

CHECK

$$\begin{aligned} 2(1) - 0 + (-1) &= 1 \quad \checkmark \\ 2(1) + 2(0) + 4(-1) &= -2 \quad \checkmark \\ 4(1) + 0 &= 4 \quad \checkmark \end{aligned}$$

Slide 54 • Back substitution

Clear above the third pivot: $R_2 \leftarrow R_2 - R_3$ gives $[0 \ 1 \ 0 \mid 0]$; $R_1 \leftarrow R_1 - \frac{1}{2}R_3$ gives $[1 \ -\frac{1}{2} \ 0 \mid 1]$. Clear above the second pivot: $R_1 \leftarrow R_1 + \frac{1}{2}R_2$ gives $[1 \ 0 \ 0 \mid 1]$. The left block is the identity and the right column is the solution: $a = 1, b = 0, c = -1$.

Check in the original equations: $2 - 0 - 1 = 1$; $2 + 0 - 4 = -2$; $4 + 0 = 4$. (Historical note for the instructor: the source material listed the third equation as $4a + b = -1$, which does not fit this walkthrough; $4a + b = 4$ is the consistent version.)

What if the system is singular?

Look at the row of zeros, then at its constant.

$$\begin{bmatrix} 1 & 2 & -1 & 5 \\ 2 & 4 & 5 & 1 \\ 3 & 6 & 4 & 6 \end{bmatrix} \xrightarrow{\text{after row reduction}} \begin{bmatrix} 1 & 2 & -1 & 5 \\ 0 & 0 & 7 & -9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$0a + 0b + 0c = 0$
Infinitely many solutions
 the constant in the zero row is 0

$$\begin{bmatrix} 1 & 2 & -1 & 5 \\ 2 & 4 & 5 & 1 \\ 3 & 6 & 4 & 10 \end{bmatrix} \xrightarrow{\text{after row reduction}} \begin{bmatrix} 1 & 2 & -1 & 5 \\ 0 & 0 & 7 & -9 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

$0a + 0b + 0c = 4$
No solutions
 the constant in the zero row is not 0

Row operations used: $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$, $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$. Only the constants column differs between the two systems.

Math for Data Science · Linear Algebra, Week 2

Slide 55 • Gaussian elimination on a singular system

Augmented matrix $[1 \ 2 \ -1 \ | \ 5; \ 2 \ 4 \ 5 \ | \ 1; \ 3 \ 6 \ 4 \ | \ 6]$. $R_2 \leftarrow R_2 - 2R_1$ gives $[0 \ 0 \ 7 \ | \ -9]$; $R_3 \leftarrow R_3 - 3R_1$ gives $[0 \ 0 \ 7 \ | \ -9]$; $R_3 \leftarrow R_3 - R_2$ gives $[0 \ 0 \ 0 \ | \ 0]$. The zero row with constant 0 reads $0 = 0$: infinitely many solutions (b is free).

Change the last constant from 6 to 10 and the same steps give $[0 \ 0 \ 0 \ | \ 4]$, that is $0 = 4$: no solution. The coefficient matrix is singular either way; the constants column decides between 'many' and 'none'. This is the matrix form of the redundant and contradictory cases from Part 1.

Gaussian elimination: the algorithm

- Create the augmented matrix**
Write $[A \ | \ b]$: one row per equation, the constants in the last column.
- Reach row echelon form**
Pivot column by column: scale the pivot row to make the pivot 1, then subtract multiples of it from the rows below.
- Reduce further or back-substitute**
Clear the entries above each pivot to reach reduced row echelon form, or solve from the last equation upwards.
- Stop at a row of zeros**
A row $0 \dots 0 \ | \ 0$ means infinitely many solutions; a row $0 \dots 0 \ | \ b$ with $b \neq 0$ means no solution. Otherwise the solution is unique: the last column.

DEFINITION

Gaussian elimination is this procedure: elementary row operations applied to the augmented matrix until it is in (reduced) row echelon form.

Data science connection. This is what `numpy.linalg.solve` and `R`'s `solve()` run under the hood (with row swaps chosen for numerical stability) whenever a model fits a linear system.

Math for Data Science · Linear Algebra, Week 2

Slide 56 • Gaussian elimination: the algorithm

Summarise the four steps: (1) form the augmented matrix; (2) pivot column by column to reach row echelon form; (3) clear above the pivots to reach reduced row echelon form, or back-substitute; (4) if a zero row appears, its constant tells you whether there are infinitely many solutions (constant 0) or none (constant non-zero); otherwise read the unique solution from the last column.

Data science connection: this is what `numpy.linalg.solve`, `scipy.linalg.lu` and `R`'s `solve()` do, with rows swapped so the largest available pivot is used (partial pivoting) for numerical stability. Fitting a linear model calls such a routine on the normal equations or an equivalent factorisation.

RECAP

WEEK 2 RECAP

The definitions to remember

Linear system

A finite set of linear equations in the same variables; a solution satisfies all of them at once.

Singular vs non-singular

Non-singular: exactly one solution. Singular: infinitely many (redundant) or none (contradictory).

Row echelon form

Zero rows at the bottom; each pivot strictly to the right of the pivot above it.

Augmented matrix

 $[A \ | \ b]$: the coefficient matrix with the constants column attached.

Equivalent systems

Same solutions. Swapping, scaling by $k \neq 0$, and adding a multiple of one equation to another keep a system equivalent.

Rank

The number of independent pieces of information; number of variables minus the dimension of the solution space; the number of pivots.

Reduced row echelon form

Row echelon form with every pivot equal to 1 and zeros above every pivot. Unique for each matrix.

Gaussian elimination

Row-reduce the augmented matrix to (reduced) row echelon form; read the solution or detect a zero row.

Math for Data Science · Linear Algebra, Week 2

Slide 57 • Recap

Read the eight definitions aloud and ask students to supply an example of each from today's slides.

Suggested exit question: give a 3×3 matrix of rank 2 and explain how you know its rank without computing a determinant.

Next week builds on this: vectors, linear combinations and the geometry behind rank and the solution space.